

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <meta name="viewport" content="width=device-width, initial-scale=1.0" />

  <title>László Fábián - Portfolio</title>

  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/highlight.js/11.12.0/styles/vs2015.min.css">

  <script
src="https://cdnjs.cloudflare.com/ajax/libs/highlight.js/11.12.0/highlight.min.js"></script>

  <link rel="stylesheet" href="styles.css">

</head>

<body>

<div class="container">

  <header>

    <h1>LÁSZLÓ FÁBIÁN</h1>

    <p class="subtitle">Software Engineering & Computer Graphics Portfolio</p>

  </header>

  <article class="project-card">

    <div class="project-header">

      <h2>W.T.T.R.C. — 3D Interactive Story Game</h2>
```

<p>A modular Unity project demonstrating custom mathematical shaders, reliable state management, and an event-driven architecture.</p>

</div>

<div class="slider-wrapper">

<button class="nav-btn prev" onclick="moveSlide('main-slider', -1)">❮</button>

<div class="slides-container" id="main-slider" data-current="0">

<div class="slide active">

<div class="module-title">State Serialization & Lifecycle Management</div>

<p class="module-desc">A centralized JSON serialization system that persists game state across sessions and manages complex object lifecycles.</p>

<div class="video-wrapper">

<iframe src="youtubelinkremovedforviewing" allowfullscreen></iframe>

</div>

<div class="ide-container">

<div class="tab-bar">

<button class="tab-btn active" onclick="switchTab(this, 's1-file1')">SaveManager.cs</button>

<button class="tab-btn" onclick="switchTab(this, 's1-file2')">Saveable.cs</button>

<button class="tab-btn" onclick="switchTab(this, 's1-file3')">SettingsManager.cs</button>

<button class="tab-btn" onclick="switchTab(this, 's1-file4')">LauncherManager.cs</button>

```
<button class="tab-btn" onclick="switchTab(this, 's1-  
file5')">PauseManager.cs</button>
```

```
</div>
```

```
<div id="s1-file1" class="file-pane active"><pre><code class="language-  
csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
<div id="s1-file2" class="file-pane"><pre><code class="language-csharp">REMOVED  
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
<div id="s1-file3" class="file-pane"><pre><code class="language-csharp">REMOVED  
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
<div id="s1-file4" class="file-pane"><pre><code class="language-csharp">REMOVED  
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
<div id="s1-file5" class="file-pane"><pre><code class="language-csharp">REMOVED  
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
</div>
```

```
</div>
```

```
<div class="slide">
```

```
<div class="module-title">Custom Motion-Vector Shaders</div>
```

```
<p class="module-desc">Screen-space shaders in HLSL built from scratch, using  
vector and matrix mathematics for real-time visual effects.</p>
```

```
<div class="video-wrapper">
```

```
<iframe src="youtubelinkremovedforviewing" allowfullscreen></iframe>
```

```
</div>
```

```
<div class="ide-container">
```

```
<div class="tab-bar">
```

```
<button class="tab-btn active" onclick="switchTab(this, 's2-  
file1')">DatamoshFeature.cs</button>
```

```
<button class="tab-btn" onclick="switchTab(this, 's2-  
file2')">DatamoshUnified.shader</button>
```

```
<button class="tab-btn" onclick="switchTab(this, 's2-  
file4')">FullscreenMoshWrapper.cs</button>
```

```
<button class="tab-btn" onclick="switchTab(this, 's2-  
file5')">DatamoshManager.cs</button>
```

```
</div>
```

```
<div id="s2-file1" class="file-pane active"><pre><code class="language-  
csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
<div id="s2-file2" class="file-pane"><pre><code class="language-hlsl">REMOVED  
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
<div id="s2-file4" class="file-pane"><pre><code class="language-  
csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
<div id="s2-file5" class="file-pane"><pre><code class="language-  
csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>
```

```
</div>
```

```
</div>
```

```
<div class="slide">
```

```
<div class="module-title">Decoupled Architecture & Event Bus</div>
```

```
<p class="module-desc">A custom event bus designed to minimize runtime  
allocations, enabling seamless communication between independent components.</p>
```

```
<div class="video-wrapper">
```

```
<iframe src="youtubelinkremovedforviewing" allowfullscreen></iframe>
```

```
</div>
```

```
<div class="ide-container">

  <div class="tab-bar">

    <button class="tab-btn active" onclick="switchTab(this, 's3-
file1')">GPUInstancedRenderer.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 's3-
file2')">ThreadSafeEventBus.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 's3-
file3')">DualSceneManager.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 's3-
file4')">InnerPlayerControl.cs</button>

  </div>

  <div id="s3-file1" class="file-pane active">

    <pre><code class="language-csharp">

REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING

    </code></pre>

  </div>

  <div id="s3-file2" class="file-pane">

    <pre><code class="language-csharp">

REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING

    </code></pre>

  </div>

  <div id="s3-file3" class="file-pane"><pre><code class="language-
csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

  <div id="s3-file4" class="file-pane"><pre><code class="language-
csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

</div>
```

</div>

<div class="slide">

<div class="module-title">Combat State Machine</div>

<p class="module-desc">A deterministic grid-based combat state machine integrated seamlessly into the core gameplay loop.</p>

<div class="video-wrapper">

<iframe src="youtubelinkremovedforviewing" allowfullscreen></iframe>

</div>

<div class="ide-container">

<div class="tab-bar">

<button class="tab-btn active" onclick="switchTab(this, 's4-file1')">MinigameManager.cs</button>

<button class="tab-btn" onclick="switchTab(this, 's4-file2')">Figure.cs</button>

<button class="tab-btn" onclick="switchTab(this, 's4-file3')">BoardSquare.cs</button>

</div>

<div id="s4-file1" class="file-pane active"><pre><code class="language-csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="s4-file2" class="file-pane"><pre><code class="language-csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="s4-file3" class="file-pane"><pre><code class="language-csharp">REMOVED CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

</div>

</div>

```
<div class="slide">
```

```
  <div class="module-title">Custom Minigames Integration</div>
```

```
  <p class="module-desc">Distinct custom minigames designed and programmed to  
interface with the overarching instance-based architecture.</p>
```

```
<div class="ide-container">
```

```
  <div class="tab-bar">
```

```
    <button class="tab-btn active" onclick="switchTab(this, 'vid-1')">Terminal</button>
```

```
    <button class="tab-btn" onclick="switchTab(this, 'vid-2')">Lockpick</button>
```

```
    <button class="tab-btn" onclick="switchTab(this, 'vid-3')">Hacking</button>
```

```
    <button class="tab-btn" onclick="switchTab(this, 'vid-4')">Circuit</button>
```

```
    <button class="tab-btn" onclick="switchTab(this, 'vid-5')">Decoding</button>
```

```
    <button class="tab-btn" onclick="switchTab(this, 'vid-6')">Calibration</button>
```

```
    <button class="tab-btn" onclick="switchTab(this, 'vid-7')">Grid_Bypass</button>
```

```
    <button class="tab-btn" onclick="switchTab(this, 'vid-8')">Override</button>
```

```
  </div>
```

```
  <div id="vid-1" class="file-pane active">
```

```
    <div class="video-tab-content"><iframe src="youtubelinkremovedforviewing"  
allowfullscreen></iframe></div>
```

```
  </div>
```

```
  <div id="vid-2" class="file-pane">
```

```
    <div class="video-tab-content"><iframe src="youtubelinkremovedforviewing"  
allowfullscreen></iframe></div>
```

```
  </div>
```

```
  <div id="vid-3" class="file-pane">
```

```
<div class="video-tab-content"><iframe src="youtubelinkremovedforviewing"
allowfullscreen></iframe></div>
```

```
</div>
```

```
<div id="vid-4" class="file-pane">
```

```
<div class="video-tab-content"><iframe src="youtubelinkremovedforviewing"
allowfullscreen></iframe></div>
```

```
</div>
```

```
<div id="vid-5" class="file-pane">
```

```
<div class="video-tab-content"><iframe src="youtubelinkremovedforviewing"
allowfullscreen></iframe></div>
```

```
</div>
```

```
<div id="vid-6" class="file-pane">
```

```
<div class="video-tab-content"><iframe src="youtubelinkremovedforviewing"
allowfullscreen></iframe></div>
```

```
</div>
```

```
<div id="vid-7" class="file-pane">
```

```
<div class="video-tab-content"><iframe src="youtubelinkremovedforviewing"
allowfullscreen></iframe></div>
```

```
</div>
```

```
<div id="vid-8" class="file-pane">
```

```
<div class="video-tab-content"><iframe src="youtubelinkremovedforviewing"
allowfullscreen></iframe></div>
```

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</div>
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</div>
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</div>
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</div>
```

```
<button class="nav-btn next" onclick="moveSlide('main-slider', 1)">&#10095;</button>
```


</div>

<div class="dots" id="main-dots">

</div>

</article>

<article class="project-card">

<div class="project-header">

<h2>Event-Driven Desktop Grid Simulation</h2>

<p>A cross-platform desktop algorithmic simulation using MVVM architecture to separate presentation and application logic.</p>

</div>

<div class="video-wrapper">

<iframe src="youtubelinkremovedforviewing" allowfullscreen></iframe>

</div>

<p style="text-align: center; margin: 1.5rem 0 0.8rem 0; color: var(--text-muted); font-size: 0.95rem;">

Core Logic & Architecture:

</p>

```

<div class="ide-container">

  <div class="tab-bar">

    <button class="tab-btn active" onclick="switchTab(this, 'eve-
file1')">MainWindow.xaml</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-
file2')">MainWindow.xaml.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-
file3')">MainViewModel.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-
file4')">ViewModelBase.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-
file5')">DelegateCommand.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-file6')">GameLogic.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-file7')">GameField.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-
file8')">GameStorage.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-file9')">GameData.cs</button>

    <button class="tab-btn" onclick="switchTab(this, 'eve-
file10')">Documentation</button>

  </div>

```

```

<div id="eve-file1" class="file-pane active"><pre><code class="language-xml">&lt;!--
<Window x:Class="Menekulj.WPF.MainWindow"

  xmlns="http://schemas.microsoft.com/winfx/2006/xaml/presentation"

  xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"

  Title="Menekülj!" Height="800" Width="900"

  Focusable="True">

```

```

<Window.Resources>

    <BooleanToVisibilityConverter x:Key="BoolToVisibility"/>

</Window.Resources>


<Window.InputBindings>

    <KeyBinding Key="W" Command="{Binding MoveCommand}"
CommandParameter="Up"/>

    <KeyBinding Key="S" Command="{Binding MoveCommand}"
CommandParameter="Down"/>

    <KeyBinding Key="A" Command="{Binding MoveCommand}"
CommandParameter="Left"/>

    <KeyBinding Key="D" Command="{Binding MoveCommand}"
CommandParameter="Right"/>

</Window.InputBindings>


<DockPanel>

    <StackPanel DockPanel.Dock="Top" Orientation="Horizontal" Margin="10">

        <Button Content="Pályaméret választás" Command="{Binding
SelectMapSizeCommand}" Margin="5"/>


        <ComboBox SelectedItem="{Binding SelectedMapSize}"

            Visibility="{Binding IsMapSizeSelectorVisible, Converter={StaticResource
BoolToVisibility}}"

            Width="80" Margin="5">

            <ComboBox.ItemsSource>

                <x:Array Type="sys:Int32" xmlns:sys="clr-
namespace:System;assembly=mscorlib">

                    <sys:Int32>11</sys:Int32>

```

```

        <sys:Int32>15</sys:Int32>

        <sys:Int32>21</sys:Int32>

    </x:Array>

</ComboBox.ItemsSource>

</ComboBox>

    <Button Content="Új játék" Command="{Binding NewGameCommand}"
Margin="5"/>

    <Button Content="{Binding PauseButtonText}" Command="{Binding
TogglePauseCommand}" Margin="5"/>

    <Button Content="Mentés" Command="{Binding SaveGameCommand}" Margin="5"
IsEnabled="{Binding CanSaveOrLoad}"/>

    <Button Content="Betöltés" Command="{Binding LoadGameCommand}" Margin="5"
IsEnabled="{Binding CanSaveOrLoad}"/>

    <TextBlock Text="{Binding TimeDisplay}" Margin="20,0,0,0"
VerticalAlignment="Center" FontWeight="Bold"/>

    <TextBlock Text="(WASD Irányítás)" Margin="10,0,0,0" VerticalAlignment="Center"
FontStyle="Italic"/>

</StackPanel>

<Border DockPanel.Dock="Bottom" Margin="10" BorderBrush="Gray"
BorderThickness="1" Background="White">

    <Canvas x:Name="GameCanvas" Focusable="True"

        Width="{Binding CanvasWidth, Mode=OneWayToSource}"

        Height="{Binding CanvasHeight, Mode=OneWayToSource}">

    </Canvas>

</Border>

```

```
</DockPanel>

</Window> --&gt;</code></pre></div>

<div id="eve-file2" class="file-pane"><pre><code class="language-csharp">REMOVED
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="eve-file3" class="file-pane"><pre><code class="language-csharp">REMOVED
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="eve-file4" class="file-pane"><pre><code class="language-csharp">REMOVED
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="eve-file5" class="file-pane"><pre><code class="language-csharp">REMOVED
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="eve-file6" class="file-pane"><pre><code class="language-csharp">REMOVED
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="eve-file7" class="file-pane"><pre><code class="language-csharp">REMOVED
CONTENTS TO SHORTEN HTML WHILE VIEWING
</code></pre></div>

<div id="eve-file8" class="file-pane"><pre><code class="language-csharp">REMOVED
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="eve-file9" class="file-pane"><pre><code class="language-csharp">REMOVED
CONTENTS TO SHORTEN HTML WHILE VIEWING</code></pre></div>

<div id="eve-file10" class="file-pane">

<iframe class="pdf-frame" src="Dokumentacio.pdf" title="Game Documentation">

    This browser does not support inline PDFs. Please download the PDF to view it: <a
href="Dokumentacio.pdf">Download PDF</a>.

</iframe>

</div>

</div>

</article>
```

```
<article class="project-card">

  <div class="project-header">

    <h2>Zero-Dependency Portfolio Platform</h2>

    <p>A self-documenting static portfolio utilizing vanilla JavaScript DOM manipulation
and custom CSS properties.</p>

  </div>

  <div class="ide-container">

    <div class="tab-bar">

      <button class="tab-btn active" onclick="switchTab(this, 'web-
file1')">index.html</button>

      <button class="tab-btn" onclick="switchTab(this, 'web-file2')">styles.css</button>

    </div>

    <div id="web-file1" class="file-pane active">

      <pre><code class="language-html">REMOVED CONTENTS TO SHORTEN HTML WHILE
VIEWING</code></pre>

    </div>

    <div id="web-file2" class="file-pane">

      <pre><code class="language-css">REMOVED CONTENTS TO SHORTEN HTML WHILE
VIEWING</code></pre>

    </div>

  </div>

</article>

</div>
```

```
<script>

hljs.highlightAll();

function updateSlider(sliderId, targetIndex) {

  const slider = document.getElementById(sliderId);

  if (!slider) return;

  const slides = slider.querySelectorAll('.slide');

  const dotsContainerId = sliderId.replace('slider', 'dots');

  const dotsContainer = document.getElementById(dotsContainerId);

  const dots = dotsContainer ? dotsContainer.querySelectorAll('.dot') : [];

  if (targetIndex >= slides.length) targetIndex = 0;

  if (targetIndex < 0) targetIndex = slides.length - 1;

  slider.dataset.current = targetIndex;

  slides.forEach((slide, i) => {

    slide.classList.toggle('active', i === targetIndex);

  });

  dots.forEach((dot, i) => {

    dot.classList.toggle('active', i === targetIndex);

  });

}
```

```
function moveSlide(sliderId, step) {  
    const slider = document.getElementById(sliderId);  
    const currentIndex = parseInt(slider.dataset.current, 10);  
    updateSlider(sliderId, currentIndex + step);  
}  
  
function setSlide(sliderId, index) {  
    updateSlider(sliderId, index);  
}  
  
function switchTab(buttonElement, targetFileId) {  
    const ideContainer = buttonElement.closest('.ide-container');  
  
    const tabs = ideContainer.querySelectorAll('.tab-btn');  
    tabs.forEach(tab => tab.classList.remove('active'));  
    buttonElement.classList.add('active');  
  
    const panes = ideContainer.querySelectorAll('.file-pane');  
    panes.forEach(pane => pane.classList.remove('active'));  
  
    const targetPane = ideContainer.querySelector(`#${targetFileId}`);  
    if (targetPane) {  
        targetPane.classList.add('active');  
    }  
}  
  
</script>
```


</body>

</html>